Guorui XIE

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EDUCATION

Tsinghua University, Department of Computer Science and Technology, *Ph.D.* 2019.09 - Now
Sun Yat-Sen University, School of Computer Science and Engineering, *B.Eng.* 2015.09 - 2019.06

RESEARCH

My main research interest is "intelligent network traffic analysis". Based on the resources and programmability of network switches/routers, 1) deploy customized machine learning models to conduct high-accuracy, low-latency traffic analysis; 2) design the real-time, high-throughput traffic feature obfuscation to prevent traffic analysis attacks; 3) develop the high-accuracy, real-time traffic measurement and scheduling system.

I published 11 papers and 8 patents, won the second prize (30K CNY) in the "Video Pedestrian Counting Challenge for Embedded Terminals" competition of 2018 China PRCV Conference, 2021 and 2022 Tsinghua-Ping An AI BANK Scholarship (8K CNY), 2023 Tsinghua-Nanhu Talent Scholarship (10K CNY), and 2023 high-level postgraduate project funding (150K CNY) from China Scholarship Council. I was a core participant in two national natural science projects and one national laboratory (Peng Cheng Lab) "SDN Controller and Advanced Programmable Environment" project.

PUBLICATION

1. Mousika: enable general in-network intelligence in programmable switches by knowledge distillation
   first author, INFOCOM 2022 (CCF A)
2. Generating neural networks for diverse networking classification tasks via hardware-aware neural architecture search
   first author, Transactions on Computers 2023 (CCF A)
3. Empowering in-network classification in programmable switches by binary decision tree and knowledge distillation
   first author, Transactions on Networking 2023 (CCF A)
4. Dryad: deploying adaptive trees on programmable switches for networking classification
   first author, ICNP 2023 (TsinghuaCS A, CCF B)
5. Efficient attack detection with multi-latency neural models on heterogeneous network devices
   first author, ICNP 2023 (TsinghuaCS A, CCF B)
6. Efficient flow recording with InheritSketch on programmable switches
   first author, ICDCS 2023 (CCF B)
7. Soter: deep learning enhanced in-network attack detection based on programmable switches
   first author, SRDS 2022 (CCF B)
8. Self-attentive deep learning method for online traffic classification and its interpretability
   first author, Elsevier Computer Networks 2021 (CCF B)
9. SAM: self-attention based deep learning method for online traffic classification
   first author, NetAI@SIGCOMM 2020 (CCF A Workshop)
10. In-Forest: distributed in-network classification with ensemble models
    third author, ICNP 2023 (TsinghuaCS A, CCF B)
11. Pontus: finding waves in data streams
    sixth author, SIGMOD 2023 (CCF A)